Literature-Implied Negative Answer for arXiv:1912.12365

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Abstract

Burton and Juschenko's arXiv:1912.12365 formulates two strengthening conjectures around positive definite extensions and proves that they imply Connes' embedding conjecture. Later literature, via the theorem $\mathrm{MIP}^* = \mathrm{RE}$, refutes Connes' embedding. Therefore the half finite approximation conjecture and the low energy extension conjecture of arXiv:1912.12365 are false by contrapositive.

Source conjectures

In arXiv:1912.12365, Burton and Juschenko state Connes' embedding conjecture as Conjecture 1.1, source label `thm.CEC`:

$$C^*(\mathbb{F}) \otimes_{\max} C^*(\mathbb{F}) = C^*(\mathbb{F}) \otimes_{\min} C^*(\mathbb{F}).$$

They then state Conjecture 1.3, source label `thm.half`, on existence of half finite approximations for representations of $\mathbb{F} \times \mathbb{F}$. They also state Conjecture 1.5, source label `lem.nogain`, on extending finite families of normalized strictly positive definite functions without increasing the pairwise energy quantities.

Implication supplied by the source paper

The source paper explicitly supplies the implication chain needed here. Its introduction says that the main topic is a theorem showing `lem.nogain` $\Rightarrow$ `thm.CEC`. In the subsection "Deducing Connes' embedding conjecture from half finite approximation conjecture", the authors prove

$$\texttt{thm.half} \Longrightarrow \texttt{thm.CEC}.$$

The proof ends by saying that, in order to prove Connes' embedding, it suffices to prove `thm.half`. Later, the section titled "Proof of Conjecture `thm.half` from Conjecture `lem.nogain`" states that the paper proves

$$\texttt{lem.nogain} \Longrightarrow \texttt{thm.half}.$$

Thus, as stated in arXiv:1912.12365,

$$\texttt{lem.nogain} \Longrightarrow \texttt{thm.half} \Longrightarrow \texttt{thm.CEC}.$$

Later literature

Ji, Natarajan, Vidick, Wright, and Yuen prove $\mathrm{MIP}^* = \mathrm{RE}$ in arXiv:2001.04383 and state that their result refutes Connes' embedding conjecture. Goldbring's survey arXiv:2109.12682 records the negative solution to the Connes embedding problem and explains its relationship to the Kirchberg/QWEP/Tsirelson equivalence web.

Conclusion

The accepted later literature gives

$$\neg\,\texttt{thm.CEC}.$$

Since arXiv:1912.12365 proves `thm.half` $\Rightarrow$ `thm.CEC`, the half finite approximation conjecture `thm.half` is false. Since the same source proves `lem.nogain` $\Rightarrow$ `thm.half`, the low energy extension conjecture `lem.nogain` is also false.

This is a literature-implied negative answer. It does not construct an explicit finite counterexample to `lem.nogain`, and it does not affect the weak extension theorem proved in the source paper.

References

[1] Peter Burton and Kate Juschenko. *Extension of positive definite functions and Connes' embedding conjecture.* arXiv:1912.12365.

[2] Zhengfeng Ji, Anand Natarajan, Thomas Vidick, John Wright, and Henry Yuen. MIP* = RE. arXiv:2001.04383.

[3] Isaac Goldbring. *The Connes Embedding Problem: A guided tour.* arXiv:2109.12682.